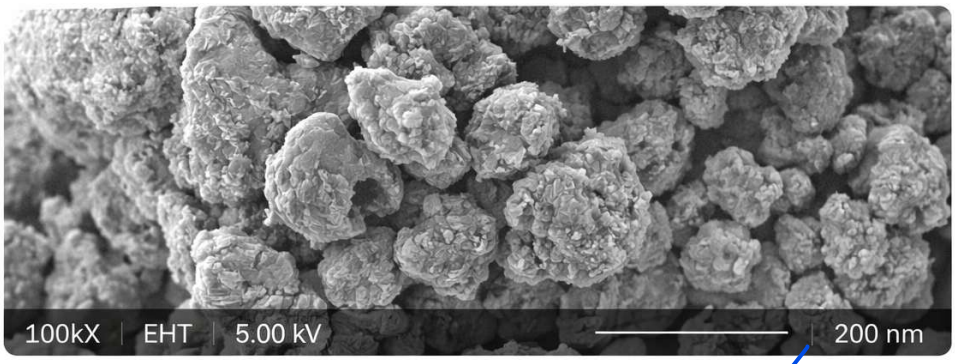
NANO LAB QUESTION BANK (B)

1. Surface Morphology Interpretation (SEM Image)

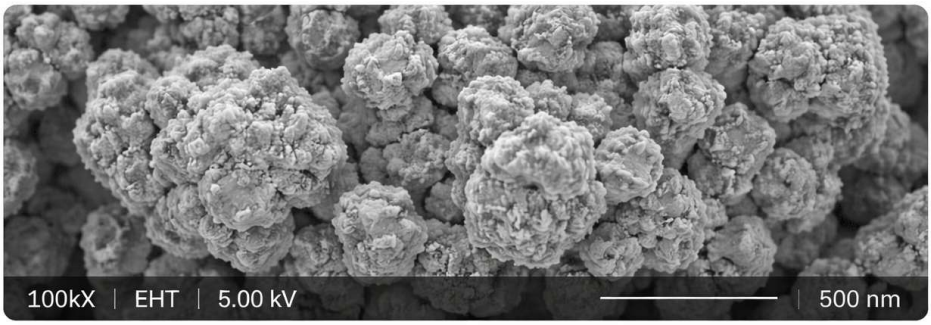
Interpret the SEM image by analyzing the following aspects: surface texture, particle shape, porosity, and uniformity. Provide scientific reasoning for each observation.



Aspect	Observation	Reason (Scientific Explanation)	Conclusion
Surface Texture	Rough and porous	Rapid nucleation and incomplete growth	Indicates non-uniform synthesis
Particle Shape	Irregular	Lack of controlled growth conditions	Suggests anisotropic growth
Porosity	Visible pores	Gas evolution or template removal	Confirms porous nanostructure
Uniformity	Non-uniform	Variation in local synthesis conditions	Indicates polydispersity
Overall Interpretation	Heterogeneous morphology	Combined growth effects	Confirms uncontrolled synthesis

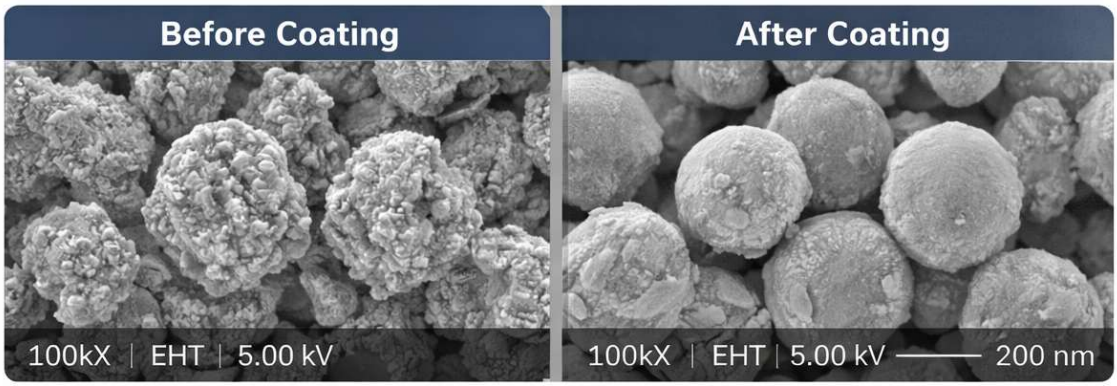
2. SEM Image: Agglomeration Analysis

An SEM image shows clusters of nanoparticles forming large aggregates. Interpret the observed morphology.



Aspect	Observation	Reason	Conclusion
Particle Distribution	Clustering observed	High surface energy → attraction	Indicates agglomeration
Interparticle Forces	Strong attraction	Van der Waals forces dominate	Weak stabilization
Surface Area	Reduced effective area	Aggregation reduces exposure	Affects reactivity
Stability	Poor dispersion	Lack of capping agents	System is unstable
Overall	Clustered morphology	Particle interaction dominates	Confirms aggregation

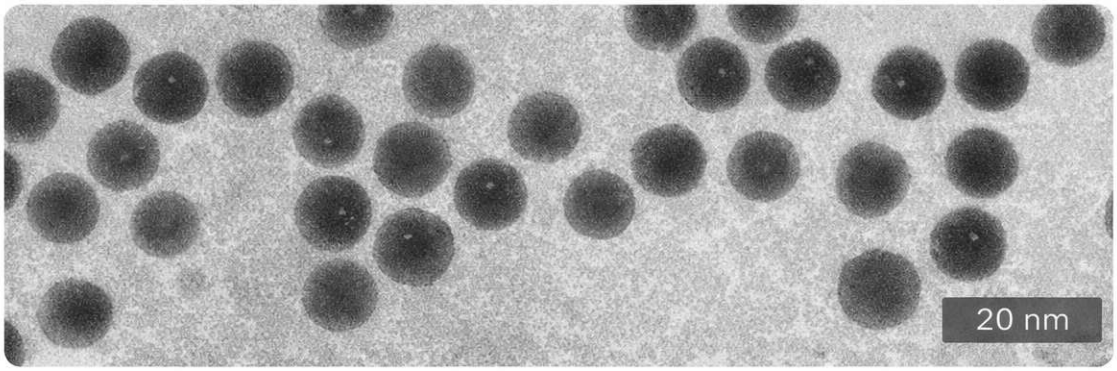
3. Two SEM images of nanoparticles are shown: one before coating and one after surface coating. Compare the images and interpret the changes in surface morphology in terms of surface smoothness, particle size, agglomeration, and contrast. Provide scientific reasoning for each observation.



Aspect	Observation	Reason	Conclusion
Surface Smoothness	Smoother after coating	Coating layer deposition	Confirms successful coating
Particle Size	Slight increase	Shell formation	Indicates core-shell structure
Agglomeration	Reduced	Surface passivation	Improves stability
Contrast	Change in brightness	Different electron interaction	Confirms material modification
Surface Energy	Before: High; After: Reduced	Coating lowers surface energy	Enhances stability

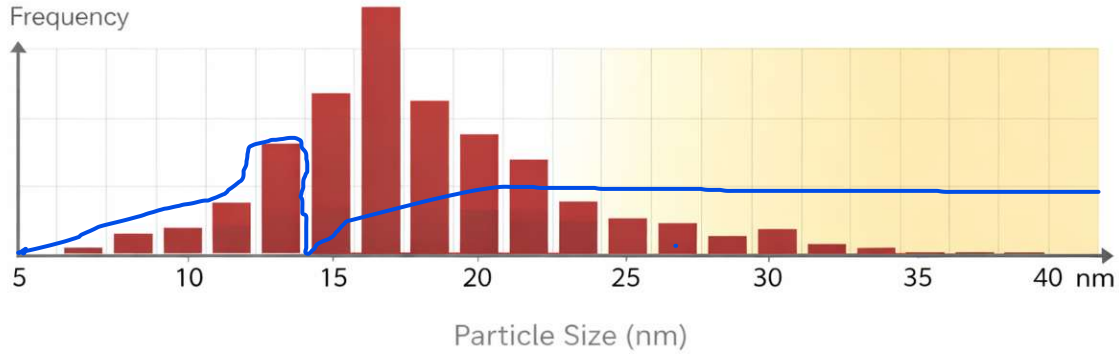
Overall	Modified morphology	Surface engineering effect	Enhanced functionality

4. A TEM image shows nanoparticles with uniform size (~10 nm). Interpret the morphology in terms of particle shape, size distribution, uniformity, and stability. Provide scientific reasoning for each observation.



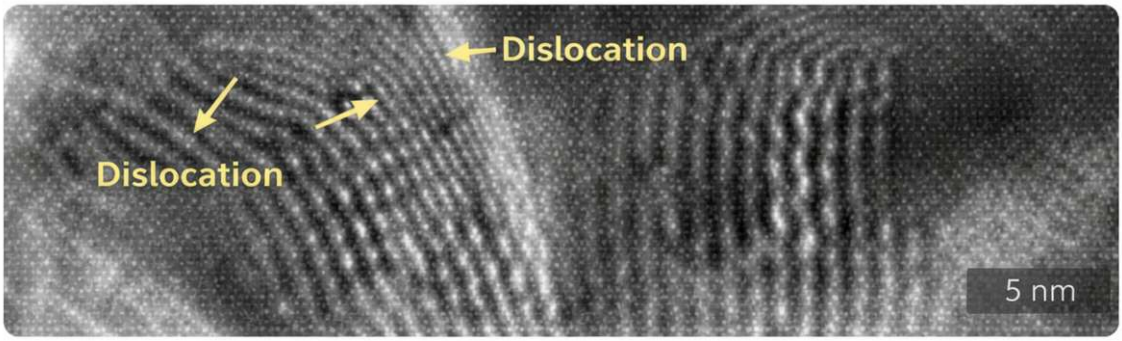
Aspect	**Observation**	**Reason (Scientific Explanation)**	**Conclusion**
Particle Shape	Spherical nanoparticles	Isotropic growth in all directions	Indicates controlled growth mechanism
Particle Size	~10 nm	Controlled nucleation and growth conditions	Confirms nanoscale size
Size Distribution	Narrow (uniform size)	Burst nucleation followed by uniform growth	Indicates monodispersed system
Uniformity	High uniformity	Consistent synthesis conditions (temperature, concentration)	Suggests controlled synthesis
Stability	Well-dispersed particles	Presence of stabilizing agents (capping/surfactants)	Indicates stable colloidal system
Agglomeration	Minimal or absent	Surface passivation prevents particle interaction	Confirms good dispersion
Contrast (TEM Image)	Dark particles on lighter background	Electron scattering by dense nanoparticles	Confirms particle visibility
Overall Interpretation	Uniform spherical nanoparticles	Controlled nucleation and growth	High-quality nanomaterial synthesis

5. A TEM histogram shows particle size distribution ranging from 5–40 nm. Interpret the data in terms of particle size distribution, nucleation process, growth uniformity, and overall synthesis control. Provide scientific reasoning for each observation.



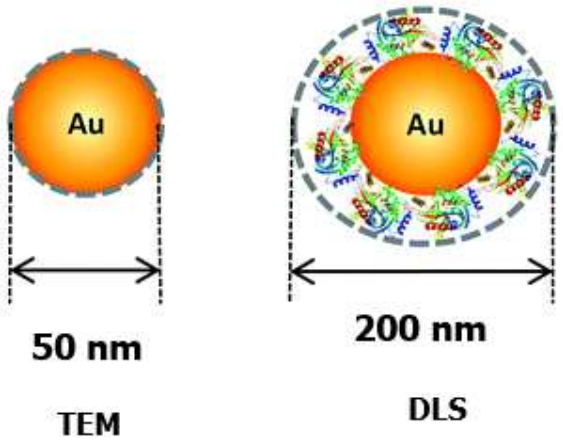
Aspect	Observation	Reason (Scientific Explanation)	Conclusion
Particle Size Distribution	Wide range (5–40 nm)	Particles formed at different times and conditions	Indicates polydispersed system
Nucleation Process	Non-uniform nucleation	Continuous or multiple nucleation events occur instead of burst nucleation	Suggests poor nucleation control
Growth Uniformity	Uneven particle growth	Different particles grow at different rates due to local variations (temperature, concentration)	Indicates non-uniform growth
Size Variation	Presence of both small and large particles	Early-formed nuclei grow larger; later ones remain smaller	Confirms time-dependent growth
Polydispersity	Broad distribution in histogram	Wide size range reflects variability in synthesis	Indicates heterogeneous system
Reaction Control	Poor control over synthesis parameters	Lack of controlled kinetics during nucleation and growth	Suggests uncontrolled synthesis
Overall Interpretation	Broad TEM histogram	Combined effect of non-uniform nucleation and uneven growth	Results in polydispersed nanoparticle system

6. A TEM image shows dislocations and lattice defects in nanoparticles. Interpret the observed features in terms of crystal structure, strain, defects, and material quality. Provide scientific reasoning for each observation.



Aspect	Observation	Reason (Scientific Explanation)	Conclusion
Crystal Structure	Distorted lattice regions visible	Presence of dislocations disrupts atomic arrangement	Indicates imperfect crystal structure
Dislocations	Line defects observed	Mismatch in atomic planes during growth	Confirms lattice defects
Strain	Local lattice distortion	Dislocations introduce internal stress	Indicates presence of strain in material
Crystallinity	Reduced in defect regions	Defects disturb periodic atomic arrangement	Suggests lower crystallinity locally
Particle Quality	Non-uniform structure	Imperfect growth conditions or rapid synthesis	Indicates reduced material quality
Growth Mechanism	Defect formation during synthesis	Fast nucleation or insufficient annealing	Suggests uncontrolled growth
Mechanical Properties	Possible weakening or strengthening	Defects influence mechanical behavior	Affects material performance
Overall Interpretation	Defective TEM image	Combined effect of dislocations and strain	Confirms presence of crystal defects

7. A nanoparticle sample shows an average particle size of ~200 nm when measured using DLS, whereas TEM analysis shows a particle size of ~50 nm. Explain the reason for this difference in measured particle size. Interpret the results in terms of measurement principle, particle environment, and aggregation effects.



Aspect	Observation	Reason (Scientific Explanation)	Conclusion
Size (DLS)	~200 nm (larger size)	Measures hydrodynamic diameter including solvent layer and surface coating	Overestimates actual size
Size (TEM)	~50 nm (smaller size)	Measures actual physical core size in dry state	True particle size
Measurement Principle	DLS: Light scattering; TEM: Direct imaging	DLS depends on Brownian motion; TEM provides real image	Different techniques give different values
Medium Effect	DLS in liquid; TEM in vacuum	Solvation layer increases apparent size in DLS	Explains size difference
Aggregation	Possible in DLS	Particle clusters scatter more light	Further increases measured size
Surface Coating	Included in DLS measurement	Capping agents increase hydrodynamic radius	Adds to size difference
Overall Interpretation	Size mismatch observed	Hydrodynamic vs actual size measurement	Techniques are complementary